Sustainable Design & Strategy

About Foodji

Foodji is a Munich-based food technology company that provides intelligent food vending machines that offer fresh, healthy, and high-quality meals. By integrating machine learning with a local supply chain, the company bridges the supply gap by providing over 100,000 meals across Germany. [1]

Sustainability is at the core of Foodji's operational areas and supply chain, involving a complex ecosystem of regional producers, corporate clients, and logistics partners. While Foodji manages a broad range of stakeholders as detailed in *Figure 1*, this report specifically targets the end-consumer and corporate decision-makers to drive sustainable behavioural change.

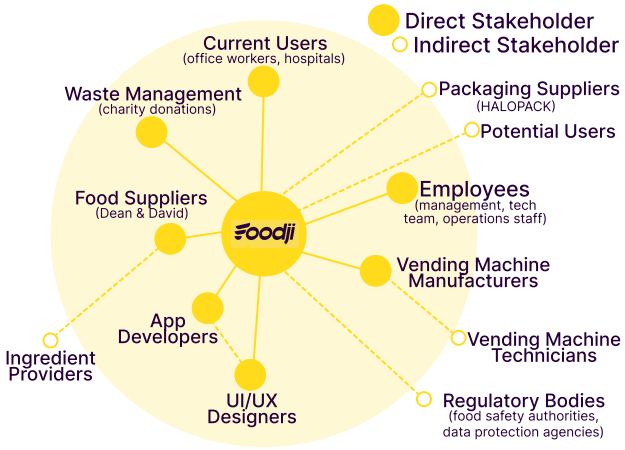


Figure 1: *Foodji's Stakeholder Map*

This project examines how Foodji can leverage its data-rich core system to implement behavioural interventions to further reduce its carbon footprint impact.

Foodji's Strengths

Data-Rich Ecosystem

Since 95% of purchases are tied to an app profile, Foodji has constant insight into consumer behaviour data, allowing for hyper-personalised targeting. [2]

AI-Driven Precision

Foodji's core strength is its proprietary algorithms that minimise waste with its demand based model, outperforming the industry average. [2]

Community Contributors

Foodji strengthens local ecosystems by partnering with regional farmers and donating surplus meals, ensuring that high-quality food supports those in need rather than going to waste. [2]

Around the Clock Availability

Foodji requires no staff on-site and provides essential nutritions to underserved environments.

Sustainability Driven

Foodji's goal was to become 100% CO_2 neutral through offsetting from 2024. [2]
Prioritising regional ingredients and manufactures in Germany and remove intermediaries, reducing the transportation carbon footprint.
Use 90% less plastic film compared to standard options. [2]
Rotating catalogue of +300 meals, 50% of which are plant-based. [3]

Foodji's Challenges

Regulatory Pressure

German legislation and EU-wide carbon reporting mandates mean Foodji must move faster on packaging innovation and Scope 3 transparency than their traditional competitors might.

Ethics Struggles with Expansion

Scaling to new regions requires significant upfront capital for hardware and a complex logistical network to maintain fresh supply chains.

Intensifying Competition

The food-tech sector in Germany is increasingly crowded, with both catering firms digitising their services and new startups entering the workplace nutrition space. [4][5]

Packaging Waste

Foodji still relies on disposable packaging. Although they are recyclable, they are still single-use. [2]

Carbon Footprint Reduction

Restocking is heavily dependent on drivers and refigerated vans, where rising fuels and logistic constraints remain a constant concern.
Foodji relies heavily on constant, stable electricity and digital infrastructure.

Competitor Analysis

	Strengths vs Foodji	Weaknesses vs Foodji
selecta — Direct Provides 24/7 automated vending machines for workplaces and public spaces offering fresh food, snacks and cold beverages. [6]	• Handles diverse offerings • Over 60 years of experience • Strong, reliable maintenance and support services	• Limited specialised freshness • Less AI-driven waste reduction and user customisation • No high-quality, fresh, specialised menu
BELLABONA — Direct A corporate catering service in Germany providing healthy, sustainably packaged daily office lunches and catering. [7]	• Daily fresh meals prepared and delivered with higher quality and taste • 100% reusable packaging • Weekly rotating menu catering for specific dietary needs	• Fixed delivery times, no 24/7 availability • Orders need to be placed ahead of time via app • Daily delivery requires coordination and logistic constraints
COMPASS GROUP | Eurest — Indirect Leading catering provider in Germany, specialising in corporate dining, staff restaurants and cafeteria management. [8]	• High-quality, diverse food, freshly cooked on site • Personalised solutions • Provides social environment with staff to manage, clean and serve	• High operating costs for personnel, infrastructure and management • Limited operating hours • requires large dedicated areas

Organisational & Industry Analysis

LEGEND: Value Captured | Value Missed | Value Destroyed | Value Opportunities

ENVIRONMENT

- AI demand forecasting reduces retail food waste by 70% compared to industry average
- Regional supplier partnerships minimise transportation distances (~9km between locations)
- 50% vegetarian/vegan catalogue reduces carbon footprint of meals served
- Existing commitment to recyclable disposable packaging
- CO_2 impact of individual meals is not currently visible to users environmental data exists but isn't surfaced at point of purchase
- Renewable energy potential for vending machines remains unexplored
- Expiry-driven waste still occurs despite forecasting
- Non-recyclable dual-material packaging creates end-of-life waste that undermines sustainability claims
- High-energy refrigeration units running 24/7 without energy offsets
- Closed-loop organic waste system (compost returned to regional farmers) via existing restock routes
- Dynamic pricing on near-expiry meals converts waste risk into consumption incentive

CUSTOMERS

- 24/7 access to fresh, healthy meals; particularly valuable for shift and night workers
- App allows advance meal reservation and contactless payment
- Digitalised meal subsidies reduce administrative burden for employers
- 95% app adoption rate indicates strong user engagement
- No nutritional transparency or CO_2 impact shown at point of purchase
- No personalised sustainability profile to reflect or reinforce individual purchasing habits
- Loyalty and rewards infrastructure not yet leveraged to drive responsible consumption behaviour
- Lack of price differentiation between high and low-CO_2 meals means sustainable choices carry no financial incentive
- Over-reliance on convenience framing risks positioning Foodji as a snack provider rather than a genuine nutrition partner
- Demand for personalised health & nutrition tracking is growing workplace food providers who surface this data build stronger daily relevance
- Gamified disposal & rewards system deepens engagement beyond the point of purchase

Value Mapping

Spread the spirit of innovation, care, and enthusiasm throughout the world and to achieve our mission of zero food waste.

SOCIETY

- Solves the "catering gap" where Foodji provides space-saving & low-cost alternative over traditional canteen
- Job creation through regional supplier partnerships and local restocking personnel
- Surplus meal donation to foodsharing communities addresses food inequality
- Foodsharing currently limited to Berlin, surplus donation not yet scaled across all locations
- There's no/minimal effort to educate users about nutrition or healthy eating beyond what's available to buy
- High prices relative to canteen alternatives could limit accessibility for lower-income employee groups
- Claims of social impact without measurable reporting risks being perceived as tokenism
- Scaling surplus donation programme nationally as Foodji expands
- Using in-app sustainability profiles to make individual social impact visible and shareable
- National expansion creates an opportunity to scale surplus donation and community impact beyond Berlin, embedding Foodji in local social infrastructure across Germany

NETWORKS

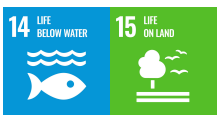
- Strong regional supplier relationships (BettaF!sh, Plantes) already embedded in operations
- Full-service model makes Foodji a reliable, low-friction partner for HR and facilities managers
- Fast installation (within 6 weeks) and consistent restocking builds trust with corporate clients
- Supplier CO_2 and related proving data not yet integrated into the customer-facing app
- No formal partnerships with waste contractors or composting facilities to close the organic waste loop
- HR, admin teams could be utilised as sustainability advocates internally
- Emerging partnerships with waste contractors, bio-gas facilities, and public health bodies could extend Foodji's value chain
- Co-branded sustainability communications with corporate clients to amplify shared impact story
- Regional supplier relationships remain underleveraged as a transparency opportunity

Figure 2: *Value Mapping*

Alignment with SDGs

Foodji has embedded the UN Sustainable Development Goals (SDGs) into the core of all their operational areas and supply chain, treating sustainability as a cost-saving and product-development strategy rather than just a slogan.

AI Planning & Packaging: Foodji uses proprietary AI to match food production with exact consumer demand at each location, reducing retail food waste. Additionally they use recyclable disposable meal packaging for simple waste separation, ensuring it is easily reintroduced into the recycling stream. Foodji actively evaluates its materials and explores future interventions.



Partners: Foodji collaborates with select regional and sustainable partners that promote meat free alternatives. Their local partners reduce transportation routes, CO_2 emissions and impact on the environment whilst supporting regional economies and communities.

Delivery Route & Filling: Foodji relies on manufacturers based in Germany and tries sourcing its ingredients regionally. Additionally, the 9 km distance between fridge and production locations ensures short transportation routes, which Foodji aims to further reduce as they expand.

Foodsharing: Foodji provides a rotating menu of over 300 nutrient-dense, fresh dishes and donating the surplus meals to charities and schemes in Berlin. By making healthy, fresh meals, Foodji empowers everyone to maintain healthy diets.

Germany Landscape

Foodji operates within the German landscape where regulatory pressure and shifts in consumer behaviour towards a circular economy mandate an integration of operational efficiency and social equity.

Politically: Germany has shifted towards high-compliance sustainability, driven by a conservative-led government and aggressive EU-wide environmental directives such as the National Strategy for Food Waste Reduction (2025-2030), which mandates a 50% per capita reduction in food waste at the retail and consumer levels by 2030. The consideration of a carbon tax on high-emissions foods like meat and dairy, further signals a regulatory environment where businesses are expected to act as primary facilitators of behavioural change. [9][10]

Socially: Germany is navigating a "Pragmatic Sustainability" phase. While the majority of Germans strive for eco-friendly, healthy lives, as evidenced by the record participation in the Pfand national water bottle deposit scheme, climateflation has made consumers highly cost-sensitive, forcing a shift towards convenience. [11] Additionally, although 59% of consumers take sustainability into account when shopping, scepticism toward opaque health and sustainability claims is at an all-time high, with 28% fof consumers feeling labels don't adequately address sustainability concerns. [12][13]

Problem Definition

Germany's food sector remains a critical barrier, accounting for over 20% of the country's total greenhouse gas emissions. While Foodji's operations touch SDG 13 and 2, SDG 12 most directly reflects Foodji's strategic position: as a delivery platform rather than producer, Foodji's primary lever is influencing consumption at the point of purchase, making demand-side behavioural change the most attributable pathway to impact.

While Foodji leads in food waste reduction through demand forecasting, there is still an intention-behaviour gap, where German consumers consider sustainability, yet their actual purchases are driven by convenience and price. Current systems fail to provide the real-time, transparent data necessary to lower information costs for sustainable choices.

Opportunity Scoping

The food sector in Germany is currently a significant environmental focal point as part of their goal to reduce overall greenhouse gas emissions by 65% by 2030, and halving food waste per capita by 2030. [14] By leveraging its AI-driven precision and sustainability-led model, Foodji is uniquely positioned to turn these regulatory and social shifts into opportunities for behavioural change.

Intervention 1: Consumer Transparency and Incentivisation

Emission transparency is a rapidly growing priority in Germany, where users strive for an eco-friendly life**.** Clear, data-driven carbon footprint labels, particularly those using colour coding, significantly shift consumer behaviour towards lower-emission meals by reducing the information cost. [15]

As an active community contributor, with a focus on sustainability, Foodji is uniquely positioned to bridge the gap between user environmental intent and actual purchase action. By integrating transparent CO_2 emission (CO_2e) labels into its digital interface, Foodji can nudge its users towards responsible consumption in exchange for tangible rewards.

Intervention 2: Dynamic Pricing for Close-to-Expiry Meals

Germany's retail sector is increasingly adopting dynamic pricing to minimise food waste, with major players like Metro demonstrating measurable reductions through expiry-based markdowns. [16] Applying a similar strategy to Foodji's system leverages existing forecasting data to push close-to-expiry meals more effectively. This not only reduces waste, but also aligns discounted meals with lower-CO_2e label purchases, reinforcing sustainability outcomes and shifts consumer preferences toward environmentally responsible choices as cost sensitivity increases.

Intervention 3: Sustainability Profiles & Leaderboard System for Low-CO_2e Meal Purchasing

Gamification has proven effective at driving behavioural change, with platforms like Duolingo and food-choice apps like Food Explorer Club demonstrating that nudging and social comparison can meaningfully influence habitual decisions. [17][18][19]

This presents an opportunity for Foodji to gamify low-CO_2e meal purchases. We propose Foodji leverages its existing customer purchase data to generate individual sustainability profiles, each featuring a running score derived directly from the CO_2e labels (A-E) of meals purchased. Accumulated points unlock tangible rewards: discounted meals, free products or local supplier vouchers, while the cumulative score feeds an opt-in leaderboard, adding social comparison to reinforce individual motivation.

Theory of Change Map

Impact Goal

By 2028, reduce the CO_2 emissions of meal production chains by 20%, by increasing monthly purchases of low-CO_2e impact meals from regular customers who are incentivised by rewards that don't exceed 2% of Foodji's revenue, tracked through in-app purchasing data.

Interventions Outline

1	2	3
Consumer Transparency & Incentivisation	Dynamic Pricing for Close-to-Expiry Meals	Sustainability Profiles & Leaderboard System for Low-CO_2e Impact Meal Purchasing

Stakeholders

- Foodji staff (operations, management, delivery drivers)
- **Foodji's app design and UI/UX team**
- Meal supplier partners
- Client companies
- Machine manufacturers

See Figure 1 for full list.

Pre-Requisites

- Meal CO_2e impact data available or calculable for all meals
- CO_2e classification criteria are defined, accurate, and credible
- Consumers already use the Foodji app for meal purchases (95% adoption)
- App infrastructure can support new features without significant disruption

• Enough of a variety of lower CO_2e food that the meals do not feel repetitive • Food that fits the consumer's personal dietary needs (balanced meal with fibre, protein, etc.) • Partners will not disengage to avoid emissions transparency	• Foodji's AI forecasting system can reliably identify meals entering the 24-hour expiry window per location • App notification infrastructure exists and is permitted by users • Health regulations permit the sale of items up to expiration	• Sufficient active consumer base to make leaderboard feel competitive and meaningful • GDPR (General Data Protection Regulation)-compliant opt-in mechanism for profile and leaderboard visibility • Local supplier partnerships confirmed for voucher reward

Assumptions

- Consumers are sustainability conscious and are willing to make changes
- CO_2e classifications are accurate enough to be credible to consumers
- Foodji's app and UI/UX team can implement required features within budget

• Consumers are happy to switch to low-CO_2e meals irrespective of their personal dietary needs • Rewards that consumers would receive are enough of an incentive to choose the low-CO_2e meal	• Foodji's AI forecasting system can reliably identify meals entering the 24-hour expiry window per location • Health regulations allow selling items up until expiration • Dynamic pricing does not cannibalise full-price sales • Discount is enough to override preference over fresh item	• Leaderboard is sufficiently motivating to change purchasing behaviour • Consumers will engage with profiles consistently over time • Local supplier partnerships remain stable and rewards remain perceived as valuable • Social comparison drives positive rather than demotivating competitive dynamics

Inputs

- CO_2e impact classification data for all meals, app infrastructure capable of purchase updates in real time, Foodji UI/UX design and development resource

• CO_2e data per meal sourced and verified • Rewards budget capped at 2% of Foodji revenue • Screen UI interface shows visible information on sustainability • In-app onboarding and educational content explaining the A-E label system • App loyalty card backend	• Foodji's existing AI demand forecasting system and expiry tracking data • Internal pricing guidelines to set discount ranges • App design updates to surface discounts clearly at point of purchase	• Existing consumers' purchase history data to generate sustainability scores • Points accumulation and reward redemption backend • Local supplier voucher agreements • Leaderboard UI design and opt-in consent flow

Activities

• Add A-E colour-coded CO_2e labels onto meal packaging • Integrate information on CO_2e per meal into the app UI at point of meal selection • Source and verify CO_2e data from meal partner producers • Build and launch monthly loyalty card: reward unlocked upon six A or B rated meal purchases within a calendar month, capped and resetting monthly • Run in-app onboarding campaign educating consumers on CO_2e label meaning & reward mechanism	• Expand AI system to trigger automatic price reductions on meals within 24 hours of expiry • Surfacing live discounts in app at point of meal selection ("reduced because it expires today") • Monitor discount uptake and adjust pricing thresholds based on waste reduction data	• Leverage existing purchasing data to generate individual sustainability profile scores, calculated by CO_2e label rating of meals purchased • Launch opt-in leaderboard with social comparison features visible in-app • Enable points accumulation redeemable against discounted meals, free products, or local supplier vouchers • Run periodic community challenges to sustain engagement over time

Outputs

• CO_2e labels live on all meal listings across the app • Loyalty card system operational and linked to A or B rated meal purchases • Monthly volume of low-CO_2e meal purchases tracked against baseline • Number of rewards unlocked and redeemed per month • Increase in purchase of sustainable meals	• Automated discount system operational for near-expiry meals • Reduction in meals discarded unsold per location per month • Number of near-expiry meals successfully sold via dynamic pricing	• Individual sustainability profiles generated and visible for all active consumers • Opt-in leaderboard live and updated in real time • Points redemption rate and reward fulfilment tracked monthly • Number of consumers achieving score improvements

Intended Outcomes

Short-Term: increased consumer awareness of meal CO_2e, initial shift in purchasing towards A or B-rated meals driven by loyalty card **Mid-Term:** habitual low-CO_2e purchasing behaviour established among regular consumers; meal partners incentivised to reduce their own emissions to gain better label ratings	**Short-Term:** reduction in expired meal waste, cost-sensitive consumers begin purchasing discounted meals **Mid-Term:** food waste reduction becomes measurable	**Short-Term:** users engage with sustainability profiles and leaderboard, initial spike in low-CO_2 meal purchases driven by competitive motivation **Mid-Term:** sustained behavioural change driven by social comparison and identity formation around sustainable eating

Long-Term: 20% reduction in CO_2e emissions across Foodji's meal production chains by 2028, behavioural change in choosing more sustainable meal options

Unintended Outcomes

• Consumers may purchase low-CO_2e meals solely for loyalty points • Meal partners may be disadvantaged or dropped from Foodji • Over-reliance on extrinsic rewards may undermine intrinsic motivation to eat sustainably	• Less food donations available in their programs • Frequent discounting may erode perceived meal quality or Foodji's brand value • Consumers may strategically delay purchases • Partners may overproduce to benefit from discount visibility	• Competitive leaderboard dynamics may demotivate consumers who fall behind • Consumers may exploit scoring system without meaningful mental shift • If rewards are removed or devalued, users who were extrinsically motivated may disengage entirely

Concept Selection

Evaluation Criteria

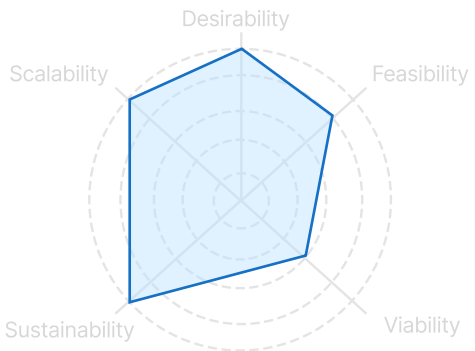
The following criteria were established which guided our decision-making to finalise an intervention. These were based off of the IDEO [20] framework, and ensured the interventions were evaluated across customer appeal, technical delivery, and financial sustainability.

Long-Term Customer Adoption (Desirability)	Assesses how effectively the intervention appeals to customer's values and motivations, and whether it sustains engagement beyond initial novelty through intrinsic or extrinsic incentives.
Technological Feasibility	Evaluates the complexity of the software and data infrastructure required, and the extent to which it can be built within Foodji's existing systems.
Return on Investment (Viability)	Compares the financial costs of implementation and ongoing operation against projected revenue impact and customer engagement goals.
Brand Loyalty (Sustainability)	Considers how well the intervention reinforces Foodji's sustainability-led brand identity and whether it strengthens long-term customer relationships.

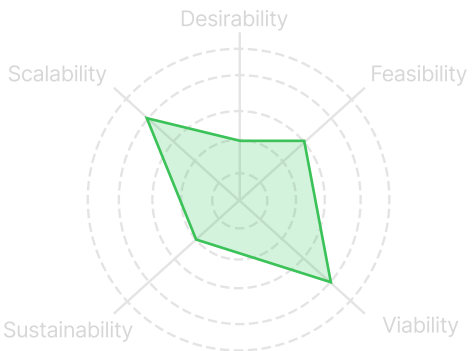
	Intervention 1	Intervention 2	Intervention 3
Long-Term Customer Adoption	Transparency appeals to eco-conscious users. However, novelty of CO_2e labelling may fade without reward iteration. Relies on consumers engaging with labels.	High initial uptake driven by cost savings. Long-term adoption depends on continued close-to-expiry stock availability; no inherent habit-forming mechanism.	Gamification with social comparison and rewards creates intrinsic and extrinsic motivation. Novelty is sustained through dynamic scoring and opt-in competition.
Technological Feasibility	Requires CO_2e data integration from partners and a label display system in-app. Partner data quality is a dependency outside Foodji's direct control.	Foodji's demand forecasting algorithm already tracks expiry data. The dynamic price adjustment is a software extension with minimal new infrastructure.	Leverages existing app purchase data, and payment systems. Scoring engine and leaderboard are software-only additions with no hardware dependency.
Return on Investment	Low implementation cost but reward spend must stay < 2% revenue. ROI is indirect via increased low-CO_2e meal sales and strengthened relationships with eco-conscious customers. Risk of losing partnerships with higher CO_2e meal providers.	Selling a meal at a discount still generates more revenue than discarding it as waste, making the financial risk low. Combined with the minimal development cost required, this intervention is likely to be revenue-neutral or better.	Low upfront cost, but ongoing reward redemption costs scale with engagement. ROI depends on sustained participation, risk of reward expense exceeding impact if engagement is superficial.
Brand Loyalty	Positions Foodji as transparent and values-led, reinforcing their SDG 12 alignment. The impact depends on whether partners can supply reliable emissions data.	Reduces waste and saves consumers money, but is perceived as operationally reactive rather than a proactive sustainability commitment. Persistent association with near-expiry meals risks undermining Foodji's positioning as providing fresh, high quality food.	Social proof and community engagement embed Foodji's sustainability values into daily behaviour. Consumers identify with Foodji's mission, strengthening brand affinity beyond individual transactions.

Evaluation

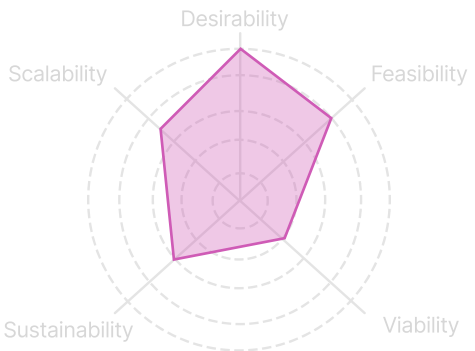
Intervention 1: Consumer Transparency and Incentivisation



Intervention 2: Dynamic Pricing for Close-to-Expiry Meals



Intervention 3: Sustainability Profiles & Leaderboard System for Low-CO_2 Impact Meal Purchasing



Decision Matrix

	Desirability	Feasibility	Viability	Sustainability
Intervention 1	High	High	Medium	High
Intervention 2	Low	Medium	High	Low
Intervention 3	High	High	Low	Medium

Final Intervention Development

The final intervention selected was **Intervention 1: Consumer Transparency and Incentivisation**. This entails two key components: firstly, CO_2 emissions are labelled for each meal, with a rating of A to E and colour-coded (green to red respectively) to easily identify the relative severity. To avoid cognitive overload, further information on emissions is integrated into the app and machine interface at the point of meal selection, rather than on the packaging itself, which will only display the letter rating and colour (*Figure 3*). As a delivery platform rather than a meal producer, Foodji will source verified emissions data from its partner producers and display this transparently to consumers.

Figure 3: *an example of showing climate score on food packaging.*

The second key component is incentivising purchasing behaviour through the addition of a monthly, digital loyalty card to Foodji's existing app (*Figure 4*). Consumers who purchase six A or B rated meals within a calendar month will unlock a reward, redeemable for discounts, free meals, or local supplier vouchers. The cap of six purchases per month keeps reward costs predictable and within the 2% revenue limit defined in the impact goal, while limiting the risk of system exploitation. The monthly reset sustains ongoing engagement by creating a recurring incentive cycle, and reflects realistic purchasing frequency: six meals per month represents an attainable threshold for regular Foodji users without penalising those with variable schedules.

Figure 4: *the digital loyalty card.*

Rationale

Intervention 1 was selected for its high desirability, feasibility, and sustainability scores, as it directly targets CO_2 emissions at the point of purchase whilst building entirely on Foodji's existing app infrastructure, making it the most immediately actionable and impactful of the three interventions.

Scalability & Adaptability

This intervention scales well because it is built entirely within Foodji's existing app and fridge screen infrastructure, meaning expansion to new cities or markets does not require additional physical adjustments, only the onboarding of new meal partners and emissions data. It is adaptable because the A-E classification system can evolve alongside improving carbon accounting methods and incoming EU regulatory labelling requirements, ensuring the intervention remains relevant and credible.

Food Production Chain

Due to Foodji's stance as a tech company, direct control over meal production emissions sits with its partner providers rather than Foodji itself. The carbon footprint of each meal is therefore shaped by decisions made upstream: ingredient sourcing, production processes, and logistics. Some partner meals draw on locally sourced ingredients, carrying a comparatively lower emissions profile, whilst others rely on ingredients shipped from further away, increasing their carbon impact. This variability makes CO_2e labelling valuable, as it surfaces differences in emissions that are otherwise invisible to the consumer at the point of purchase, and creates indirect pressure on meal partners to favour lower-emission sourcing.

Behavioural Theory Underpinning

This intervention is grounded in established behavioural frameworks. First, Nudge Theory [21] posits that presenting information at the moment of decision, without restricting choice, reliably shifts behaviour towards more sustainable outcomes. CO_2e labels reduce the information cost of making an environmentally responsible meal choice, surfacing emissions data that users would otherwise have no visibility of at the point of purchase. Further, operant conditioning principles [22] underpin the rewards mechanism: positive reinforcement through tangible rewards establishes and strengthens the association between low-CO_2e purchasing and a desirable outcome, encouraging habit formation over time rather than one-off behavioural spikes.

Alignment with Germany Landscape

The intervention aligns with the cultural and regulatory shift towards sustainability. Germany's proposed carbon tax on high-emission foods such as meat and dairy would, if introduced, directly increase the cost of the meals that Foodji's CO_2e labels would rate lowest, meaning the labelling system positions Foodji ahead of an anticipated market shift rather than reacting to it after the fact. [10]

Iteration Analysis

Element	Action	Rationale
Colour-coded CO_2e A to E labels on the meal packaging	Add	Revealing the environmental impact of the meal allows us to nudge consumers' choices without restricting their options. It encourages personal accountability and re-designs their experience to promote a sustainable outcome of purchasing low-CO_2e meals. By including the detailed information on the app and fridge screen, and only a colour on the packaging itself, the cognitive load is reduced for the consumer.
Digital loyalty card	Add	Carbon labelling alone is insufficient to drive sustained behaviour change. The digital loyalty card provides a structured, visual incentive mechanism that reinforces low-low-CO_2e purchasing beyond initial novelty.
Rewards & discounts	Add	The rewards supplement the loyalty card and bridge intention-behaviour gap by attaching tangible value to sustainable choices, that transparency alone fails to close.
Barcode scanner	Modify	Fridge purchases remain untracked. Modifying the purchase tracking mechanism to include a barcode scanner ensures all purchases are captured, extending the intervention's reach.

Expected Consequences

The primary expected outcomes from this iteration are an increase in purchases of low-CO_2e meals, and the increase of habitual sustainability-led purchasing behaviour over time. This will be achieved through tangible rewards and incentivisation , reinforcing sustainable choices without restricting user autonomy.

Considerations

The following considerations outline the practical delivery of the intervention, covering how it reaches consumers, what it requires to operate, and how its impact will be tracked.

Delivery Method

CO_2e labels (A-E and colour-coded) are added to meal packaging, and integrated directly into Foodji's app interface at the point of meal selection. Rewards are applied to users' accounts upon purchasing six low-CO_2e rated meals in a month, tracked through in-app purchasing data.

Target Audience & Affected Stakeholders

Primary: Foodji consumers who order regularly, particularly environmentally-conscious and tech-savvy consumers.
Secondary: Foodji's meal partners and producers, who will need to supply emissions data.

Inputs/Costs/Resources

Infrastructure: Uses existing app platform, new packaging labels and loyalty card component
Data: Foodji's meal providers will need to supply emissions data
Workers: UX/product team, data analysts, sustainability lead, account managers
Expenses: App development, carbon data licensing, rewards cost (capped at 2% revenue), marketing
Knowledge: Behavioural economics principles (nudge theory), purchasing data, carbon emissions per meal

Monitoring & Evaluation

Primary metric: Monthly volume of low-CO_2e (A or B rated) meal purchases versus baseline
Secondary metric: Overall CO_2e per meal ordered, reward redemption rate and user engagement with intervention
Data source: In-app purchasing data
Review cadence: Monthly dashboard review, quarterly iteration based on reward thresholds and label design
Flexibility: Reward values and label weighting can be adjusted based on uptake data

Risks

1. Consumers ignore labels → mitigate with prominent placing and onboarding education
2. Cost of rewards exceeds 2% revenue cap if uptake is high → build in dynamic scaling of reward values
3. Partner data is inaccurate or self-reported → require third-party verification
4. Consumers revert when novelty wears off → introduce tiered/streak-based rewards to maintain engagement
5. Regulatory changes to carbon labelling standards require app rework

Implementation Plan

Timeline: 2-year rollout to 2028, beginning with a data collection and onboarding phase, followed by a pilot launch, updated app rollout, and ongoing iteration based on purchase data
Roles: Enabler (Foodji product team), consumer (app users purchasing meals), partners (meal producers supplying emissions data)
Contingency: If label engagement is low, reward values will be increased. If partner data quality is poor, third-party verification will be introduced.

While these risks were considered, the intervention's low implementation cost, adaptability, and alignment with Foodji's existing infrastructure wre deemed sufficient to outweigh them, supporting the decision to proceed.

Impact Indicators

Intervention 1: Consumer Transparency and Incentivisation

Building on the ToC map, the following indicators were considered to assess the effectiveness of Intervention 1. The screening process focused on selecting indicators that are directly traceable to the ToC's key outputs and intended outcomes.

	Indicator 1 Reduction in average CO_2e per meal	**Indicator 2** Shift in responsible meal purchases	**Indicator 3** Reward engagement & redemption rate	**Indicator 4** Consumer perception of CO_2e transparency
Type	**Environmental Indicator**	**Behavioural Indicator**	**Gamification Indicator**	**Experiential Indicator**
Relevance	**Yes.** Indicator directly measures the core long-term environmental outcome, tracking whether changes in purchasing composition are reducing CO_2e.	**Yes.** It directly measures whether CO_2e labelling and rewards are influencing purchasing decisions at the point of sale, which is the central mechanism of the intervention.	**Yes.** Reward mechanic functions as the behavioural incentive bridging CO_2e label visibility. Redemption rate indicates whether the incentive is sufficiently compelling to motivate repeated purchasing over time.	**Yes.** Assuming users are sustainability-conscious to act on environmental information depends on the CO_2e label being understood & trusted. Unclear labelling, undermines intervention's core mechanism.
Feasible	**Yes.** Supplier CO_2e data is increasingly available and will be required under PPWR regulations. Implementation cost is low once CO_2e classification is integrated into the existing app infrastructure.	**Yes.** Purchase data is already collected through Foodji's existing app. New data collection infrastructure isn't required, tracking is extension of existing point-of-sale filtered by meal CO_2e classification.	**Yes.** The reward tracking mechanism operates within the app loyalty infra- structure. The primary ongoing cost is reward expenditure, which can be monitored against the uplift in responsible meal sales to evaluate ROI.	**Yes.** In-app surveys and periodic focus groups require minimal investment. Qualitative research is cost-effective at pilot scale and can expand as the intervention rolls out.
Attributable	**Yes.** A measurable reduction in average CO_2 per meal sold following label rollout, absent from non-pilot locations, can be attributed to the change in purchasing composition driven by CO_2e label visibility.	**Yes.** comparison between pilot & non-pilot locations enables attribution of any shift in responsible meal purchasing to the CO_2e labelling and rewards mechanic, rather than to external factors.	**Yes.** Reward points are issued exclusively upon the purchase of a responsible meal, ensuring that redemption rate serves as a direct measure of whether the incentive is sustaining the intended behaviour.	**Yes.** Survey instruments can be designed to isolate the CO_2e label's specific contribution to perception change, distinguishing it from the influence of other app features or external factors.
Comparable	**Yes, internally**. No direct vending-specific industry benchmark currently exists, so reduction is compared against Foodji's own pre-intervention baseline and broader workplace catering CO_2e averages.	**Yes, internally.** No direct benchmark exists for responsible meal adoption in vending contexts. Internal pre/post comparison serves as the primary benchmark.	**Partially.** Benchmarkable against published engagement data from app-based loyalty programmes in food service and retail contexts.	**Yes, internally.** Benchmarked against Foodji's existing app satisfaction. Externally comparable against research on sustainability label comprehension, trust, & behavioural influence in digital food spaces.
Measurable	**Yes.** CO_2e per meal is calculable using supplier-provided emissions data, transportation figures & app purchasing records. Baseline: weighted average CO_2e per meal sold per location prior to intervention launch.	**Yes.** The proportion of responsible meals purchased to total meals sold, directly trackable through in-app point-of-sale data. Baseline: % of responsible meals purchased per location during the pre-launch period.	**Yes.** Points issued and redeemed are directly trackable through the app's loyalty infrastructure. Baseline: user engagement with any existing app loyalty features prior to the launch of the new reward mechanic.	**Yes.** Consumer perception is measurable using a combination of quantitative survey & qualitative research methods. Baseline: a pre-launch survey capturing users' current satisfaction with food choice information & app usability.
Data Collection Method	Supplier-provided CO_2e data matched against in-app purchase records per location. Monitored monthly across pilot and non-pilot locations.	In-app point-of-sale records tracking responsible versus non-responsible meal purchases per location per week, monitored against the pre-launch baseline.	App backend data tracking points issued per responsible meal purchased, redemption rate per user, and repeat purchase frequency among reward users.	Quarterly in-app post-purchase survey prompts assessing label clarity, trust, and reported influence on purchase decision.

Assessing impact of outcomes

After evaluating the impact indicators, the main metric to guide our outcome assessment is **Indicator 1**, focused on reducing CO_2e by shifting employees' meal consumption from high to low emission meals through transparent CO_2e labelling and incentivisation. The following assumptions were used:

- Foodji serves 100,000 meals to employees per month across all their Germany locations [1]
- CO_2e per meal per person in Germany is estimated to be 2300kg CO_2e, which equates to 2.1 kg CO_2e per meal, however Foodji's baseline is estimated at 1.9 kg CO_2e per meal, naturally falling below the German national average due to their multiple sustainability measures. [23]
- To achieve 20% reduction of CO_2e by 2028, the average intensity must drop to 1.52kg CO_2e per meal
- Meals CO_2e factor was calculated using the HelloFresh methodology in partnership with HowGood with the following equation used: [24] (Additional calculations can be seen in the Appendix)

*Total Meal CO_2e = (Ingredient Weight * Emission Factor) + Cooking + Packaging + Logistics + Storage*

- Assumed percentage of consumed volume per Foodji's meal type and calculated target volume:

Target Volume Equation:
$(x * 0.6)+(y*1.2)+(z*3)=1.44$ ⟶
$x=0.35$, $y= 1-x-z$, $z=1-z-y$

Meal category	Emissions Factor (CO_2e)	Baseline Volume(Actual)	Target Volume(After Intervention)
Low (A/B rating)	0.6kg	15%	35%
Medium (C rating)	1.2kg	45%	45%
High (D/E rating)	3.0kg	40%	20%

Outcome 1: Successful CO_2e Reduction per Meal

- **Type of Outcome:** Environmental & Behavioural Shift
- **Indicator:** Indicators 1 & 2
- **Diversity:** Environmental data (CO_2e) calculation & behavioural adoption rates
- **Time Frame:** 24 months

Data Measurement

- App & vending machine analytics tracking consumer engagement with CO_2e labels and decision changes
- Loyalty points redemption rates indicating effectiveness of incentives

Uncertainty: Medium

- Data collection is reliable
- Consumer response affects adoption rate
- Initial implementation might show positive interest, but novelty could wear off and recurring change in user trends are harder to achieve

Feasibility

1. CO_2e data is calculable from supplier ingredients, and purchase data is already tracked via the app.
2. Loyalty points for purchases can be integrated into Foodji's app.
3. CO_2e labels will be implemented on existing Foodji platforms (website, app, vending machines) to shift consumer behaviour

Impact: High

- Directly linked to intervention success
- Directly achieves 25% reduction of CO_2e, exceeding 20% impact goal, with a 5% buffer

CO_2 Emissions Reduction

- Baseline meals produced per month = 100,000
- CO_2e total meals equation = *(Volume_low_meals% x CO_2e_low)+ (Volume_mid_meals% x CO_2e_mid) + (Volume_high_meals% x CO_2e_high)*
- Baseline CO_2e from meals per month = (15%x0.6)+(45%x1.2)+(40%x3.0) = 183,000 CO_2e/month
- Post intervention= (35%x0.6)+(45%x1.2)+(20%x3.0) = 135,000 CO_2e/month
- **Total emissions reduction** = Baseline emissions - Post-intervention emissions = 48,000 kg CO_2e saved/month → 576 Tonnes CO_2e saved/year

Outcome 2: Economic Viability of Incentives

- **Type of Outcome:** Financial
- **Indicator:** Indicators 2 & 3
- **Diversity:** Measures fiscal expenditure & user loyalty engagement
- **Time Frame:** Ongoing (monthly reset)

Data Measurement

- App & vending machine analytics tracking consumer engagement and user purchase history
- Loyalty points redemption rates indicate effectiveness of incentives

Uncertainty: Medium

- Costs are strictly capped at 2% revenue limit
- Consumer perception of incentive
- Length and amount of available incentives

Feasibility

1. All points issued & redeemed can be directly tracked through Foodji's existing app
2. Limited UI edits and implementation, simple A-E colour coded labels with available extra CO_2e details
3. Loyalty points are only achievable through meal purchases to achieve genuine long term customer diet shift and avoid system 'cheat' through purely snack purchases

Impact: Medium-High

- Bridging price barrier is essential for cost-sensitive German customer, whilst remaining beneficial and profitable for Foodji
- Urgency drives rapid adoption

Estimated Cost of Incentives

- Foodji annual revenue: €5,000,000 [25]
- Capped 2% annual expenditure = €100,000
- Capped 2% monthly expenditure = €100,000/12 = €8,333
- 50% meal discount voucher = €7(per meal)/2 = €3.50
- Foodji consumers served per month = $\frac{100{,}000}{4(\text{weeks})\text{x}5(\text{days})}$ = 5,000
- Free meals available per month = $\frac{\textit{Monthly Expenditure}}{\textit{Discount Cost}} = \frac{8333}{7}$ = 1190 meals (~20%)
- Discounted meals available per month = $\frac{\textit{Monthly Expenditure}}{\textit{Meal Cost}} = \frac{8333}{3.50}$ = 2380 meals
- Snack rewards available per month = $\frac{\textit{Monthly Expenditure}}{\text{Snack Cost}} = \frac{8333}{2.50}$ = 3333 snacks
- Given a company subsidises a meal's price for their employees, Foodji can afford to give 1 free meal to 47% of its total user base per month, as opposed to ~20%
- Effective discount for user spending = $\frac{140/\text{month (20 meals)}}{7}$ = 5% highly competitive

Outcome 3: User Retention, Acquisition & Lifetime Value

- **Type of Outcome:** Economic/Brand Equity
- **Indicator:** Indicators
- **Diversity:** Quanlitative user feedback & Quantitative conversion data
- **Time Frame:** Review at 24 months, ongoing

Data Measurement

- App & vending machine analytics tracking click through rate for all consumer types
- Targeted 'CO_2e Transparent' meal satisfaction: 4.5/5 star rating

Uncertainty: Medium

- Data is internal and highly controllable and easily analysed through app.
- Perception is subjective and can be influenced by external factors.

Feasibility

1. All metrics are captured natively within the Foodji app's user analytics.
2. Simple post-purchase meals satisfaction
3. Bi-yearly sustainable perception polls

Impact: High

- Successfully mainstreams sustainability turning it into standard

Estimated User Retention

- Increase of 'CO_2e Transparency' driven purchases: from 10% to 40% in 24 months
- Engagement necessary shift: 25-35%, approx 1250-1750 consumers
- Reduced overall monthly churn by 5%
- 5,000 consumers x 5%= 250 consumers/month enagaged
- Protected revenue = 250 x €140 avg. spend = €35,000
- Acquisition Calculation assuming 5% referral rate: 1,250 consumers x 5% referral rate = 63 new consumers/month

Strategy Implementation Roadmap

Phase 1: Pilot Launch

- Launch of CO_2e labels and loyalty card to pilot locations in Munich
- In-app onboarding campaign live, educating consumers on A-E labelling system and reward mechanism
- Barcode scanner operational for purchases outside the app
- Data collection tracks volume of low-CO_2e meal purchases per location per week

~€2000 Marketing & onboarding campaign

~€100.000 Reward fulfilment (2% of revenue)

Phase 3: Full Expansion

- Full rollout across all Foodji locations in Germany
- Meal partners actively improving emissions profiles to achieve better label ratings
- Habitual use established

~€100.000 Ongoing reward fulfilment

~€10.000 National rollout marketing & partner communications

~€0.05 per unit Addition of A-E labels on packaging

2026

Pre-Implementation Phase

- CO_2 emissions data collection and verification from meal providers
- A-E classification methodology defined
- A-E classification printed onto packaging
- App development: CO_2e information and labelling, loyalty card, barcode scanner integration
- Finalise reward offerings within Foodji and with local suppliers
- Design onboarding campaign
- Staff training for Foodji operations team

~€8.000 App development

~€0.05 per unit Addition of A-E labels on packaging

~€2000 Carbon data licensing & verification

~€5000 Staff training

~€10.000 Marketing

2027

Phase 2: Review and Iteration

- Monthly dashboard review of low-CO_2e purchase volumes
- Quarterly iteration on reward thresholds and label placement based on user uptake
- Assess partner data quality → introduce third-party verification if needed
- Evaluate barcode scanner adoption rate → adjust if engagement is low

~€100.000 Ongoing reward fulfilment

~€2000 Third-party data verification (if required)

~€10.000 Potential re-design of barcode scanner based on evaluation

2028

Phase 4: Impact Measurement

- Continuous tracking of low-CO_2e meal purchase volumes across all locations
- Annual review of aggregate CO_2 emissions per meal against pre-intervention baseline
- Impact goal assessed

~€100.000 Ongoing reward fulfilment

~€3.000 Annual third-party CO_2e audit & impact verification

Risks and Mitigating Strategies

	Risks	Impact	Mitigation Strategies
Consumers	1. Users may not prioritise tracking their CO_2e impact if it feels like an added effort on top of a convenient service. The core value proposition of Foodji is ease and speed, so any friction introduced by the labelling system risks being ignored altogether. 2. Users may also perceive the rewards as insufficient, relative to the conscious effort of selecting a lower-CO_2e meal every time, defaulting back to habit or taste-based choices instead. 3. Additionally, a smaller subset of users may game the reward system by purchasing responsible meals solely for points without genuine behaviour change, or disengage entirely once the novelty wears off.	The core behaviour change mechanism fails to gain traction, undermining the intervention's outputs.	1. Ensure the CO_2e label is embedded seamlessly into the existing purchase journey rather than as a separate feature, minimising perceived effort. 2. Conduct user research during the pilot to assess reward appeal and adjust incentive value accordingly. 3. Implement a monthly reward cap to manage gaming risk while maintaining genuine engagement.
Operations	1. CO_2e classification data per meal may be inaccurate or inconsistent if supplier emissions data is unreliable or incomplete. 2. Defining a CO_2e threshold may be contested internally. 3. Uptake of the intervention is higher than expected → may lead to financial strain for Foodji, or surpass the allocated budget.	Inaccurate classifications undermine trust and the credibility of the entire intervention. Differentiating or conflicting signals may risk confusing users at the point of purchase.	1. Establish a supplier data verification process and audit CO_2e classifications periodically. 2. Define responsible meal criteria early and lock in before launch. 3. Define pilot locations in Munich before full rollout.
Suppliers	1. The app infrastructure may not support real-time CO_2e label updates or reward point synchronisation at scale across all locations simultaneously. 2. CO_2e classification algorithm may require significant development resource, delaying launch.	Resistance in suppliers may delay the meal deliveries to Foodji stations. Classification of meals on some levels may risk damaging Foodji's regional supplier relationships which are a core part of their proposition.	1. Explore digital-only CO_2e labelling via app as primary channel, reducing dependency on physical packaging changes. 2. Communicate classification criteria transparently to suppliers ahead of rollout and provide transition support where required, especially in the initial stages.
Technical	1. The app infrastructure may not support real-time CO_2e label updates or reward point synchronisation at scale across all locations simultaneously. 2. CO_2e classification algorithm may require significant development resource, delaying launch.	Technical delays push back the intervention timeline.	1. Phase the rollout, pilot at high-density locations first before scaling. 2. Conduct a technical feasibility assessment early with the app design and UI/UX team.

Intended and Unintended Outcomes

	Intended Outcomes	Unintended Outcomes
Short-Term (Year 1) Implementation & Adoption	• Successful integration of A-E CO_2e labels for 100% of the active catalogue. • Increased customer awareness with verified CO_2e transparency. • Measurable reduction of high CO_2e meal consumption.	• Segment of consumers may find labelling and carbon data overwhelming at the beginning. • Sudden, sustained demand for A/B-rated meals may lead to stock-outs of popular vegan/vegetarian items. • Potential friction if monthly reward is reached too early.
Mid-Term (Years 1-2) Habituation & Profit Recovery	• High-frequency consumers internalise CO_2e ratings, leading to sustained CO_2e reductions. • Spread and of regulatory measures in Germany could accelerate and ease calculation and implementation of CO_2e labels. • Consumers are successfully incentivised .	• Rise of new competitors in food-tech sector adopting similar strategies. • Sustainability behaviour may become tied to the reward • Reduction in partnerships with suppliers of higher CO_2e rated meals.
Long-Term (Years 2+) System's Transformation	• Permanent shift in consumption patterns, reducing Foodji's annual CO_2e. • Regional suppliers proactively reformulate recipes to achieve higher rating to remain visible and competitive • Labelling systems become standard in Germany • Sustainability profiles integrate into user experience, evolving into a primary loyalty metric for consumers.	• Consumers may justify higher CO_2e choices in other areas of their lifestyle. • Long-term exposure to CO_2e labels may result in consumers ignoring data as it becomes the standard. • Cost of maintaining high-fidelity CO_2e data could become a burden if simpler, industry-wide standards emerge.

Competitor Analysis

Carbon labelling and reward-based incentives have demonstrated measurable behavioural impact in comparable contexts.

	HELLO FRESH	TOO GOOD TO GO	PET CYCLE
Overview	HelloFresh launched carbon labelling in Germany and Austria in November 2021, expanding to 11 further countries in 2022 after reported success [26]. However, despite the initiative, HelloFresh's absolute CO_2 emissions have since increased, suggesting that transparency alone is insufficient to drive meaningful change at scale.	Too Good to Go rewards consumers with discounted surplus meals, embedding sustainability directly into the app's core experience. This drives habitual engagement rather than one-off behavioural change, having saved over 200 million meals globally. [30]	Germany's Pfand deposit-return system demonstrates that financial incentives embedded into everyday purchasing reliably drive long-term behaviour change at scale, achieving a return rate of 97-99% on non-reusable containers. [31]
Investment	**Moderate.** Required commissioning an ISO-compliant LCA [27] study and adding carbon labels to all packaging across multiple markets.	**Low-Moderate.** Primarily digital infrastructure, consumer rewards are built into the business model.	**High.** Significant operational infrastructure required for collection, sorting, and returns processing.
Success	**Low-Moderate.** Carbon labelling raised awareness but did not translate into measurable emissions reduction [28]. The impact was also dependent on whether consumers already held pro-sustainability values [29].	**High.** Habitual, repeat engagement rather than one-off behaviour change, saving 79 million meals in 2022 alone. [30].	**High.** Near-universal adoption in Germany [31], demonstrating that financial incentives reliably shift consumer behaviour when integrated into routine purchasing.
Comparison	The HelloFresh case demonstrates that carbon labelling alone is insufficient to drive meaningful change, whilst Too Good to Go and the Pfand system confirm that reward-based incentives produce sustained behavioural shifts in the German market. Our intervention directly addresses this gap by combining both mechanisms, requiring significantly less infrastructure and financial investment than any of the above initiatives: we avoid the cost of an ISO-compliant LCA study (€50,000-€100,000) [32] and the expenditure required to establish physical returns infrastructure for the Pfand system.		

Conclusion

The proposed Consumer Transparency and Incentivisation intervention is a financially sustainable and immediately actionable solution, building on Foodji's existing app infrastructure, and integrating minor changes to their packaging. By combining CO_2e labelling with tangible incentives, it directly addresses the intention-behaviour gap that carbon labelling alone fails to close, leveraging nudge theory and operant conditioning principles to ensure the intervention drives habit formation. With rewards costs capped at 2% of revenue and development built on the existing app and packaging, the financial risk is minimal. Unlike comparable initiatives, behavioural change is achieved at a fraction of the investment, paving the way for stronger brand-customer relations. Germany's incoming carbon tax on high-emission foods and EU-wide carbon labelling requirements mean this intervention positions Foodji ahead of regulatory change. This low-risk, high-impact and robust intervention should move into development immediately, with a pilot launching in 2026 establishing the foundation for full national rollout and a measurable 20% reduction in Foodji's CO_2 emissions by 2028.

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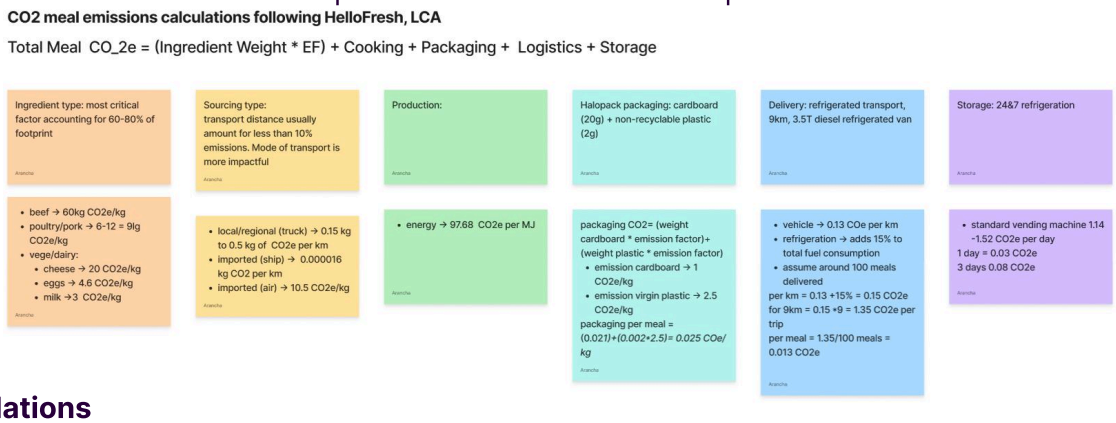
Appendix

Calculations of individual meal CO_2e

To justify the environmental impact of the Consumer Transparency & Incentivisation intervention, we utilized a Life Cycle Assessment (LCA) methodology based on the HelloFresh framework to calculate CO_2e values for individual meal components. The total meal footprint is derived from the formula:



Meal Calculations

Using the ingredient emission factors (EF) provided, we can justify the tiered meal ratings that form our baseline and target averages:

- A-Rated Meal (Vegan Bowl): Consisting of 450g of low-impact grains and vegetables
  - EF: €0.90 kg
  - Calculation: (0.45 kg x 0.9) + 0.167 = 0.572 kg CO_2e
  - Fits the < 0.6 kg "Low-Impact" threshold
- C-Rated Meal (Vegetarian Pasta): Consisting of 400g grains and 40g of high-impact cheese
  - EF: €23 kg
  - Calculation: (0.4kg x 0.9) + (0.04 kg x 20) + 0.167 = 1.197 kg CO_2e
  - Fits the ~1.2 kg "Medium-Impact" threshold
- D/E-Rated Meal (Beef Dish): Consisting of 400g vegetables and a 45g portion of beef
  - EF: €60 kg
  - Calculation: (0.4 kg x 0.8) + (0.045 kg x 60) + 0.167 = 3.227 kg CO_2e
  - Fits the > 3.0 kg "High-Impact" threshold

Target Volume Calculations

1. The primary driver for these specific numbers is the requirement to drop the average meal footprint from 1.83 kg to a level that satisfies Foodji's 2028 sustainability goals.
   - Using your equation (x x 0.6) + (y x 1.2) + (z x 3.0) = 1.35, setting x to 0.35 and z to 0.20 is the most efficient way to achieve a 26.2% reduction, obtaining the 20% impact goal with 5% buffer.
   - Sensitivity: If x were lower (e.g., 25%), the average would rise above the target threshold, failing the intervention's core environmental KPI.
2. The shift from 40% to 20% for z (D/E meals) reflects the "Flexitarian" trend in the German market.
   - Research into consumer "nudging" suggests that roughly 50% of meat-eaters are willing to switch to a high-quality vegetarian alternative if incentivized.

- We assume 20% of the volume represents unobtainable meat consumers who will not shift regardless of incentives. Therefore, reducing z from 40% to 20% captures the entire "movable" segment of the population without making unrealistic assumptions about total behavioural overhaul.

Baseline Vs Intervention Volume Calculations

The Baseline (Current) volume assumes Foodji's current high-frequency consumption pattern, where approximately 40% of purchases remain in the High-Impact (D/E) category due to a lack of transparency. This results in a weighted average of 1.83 kg CO_2e per meal.

- Weighted Average Calculation: (0.15 × 0.6) + (0.45 × 1.2) + (0.40 × 3.0) = 1.83 kg CO_2e per meal

The intervention shifts 20% of the total volume from the high-impact tier (D/E) to the low-impact tier (A/B) through transparency labeling and the 6-meal reward incentive.

- Weighted Average Calculation: (0.35 × 0.6) + (0.45 × 1.2) + (0.20 × 3.0) = 1.35 kg CO_2e per meal